

ATPL EASA STUDY GUIDE | 022 - AGK INSTRUMENTATION

Chapter 5: Terrestrial Magnetism & Direct-Reading Compass

Syllabus Source: CAE Oxford (pp. 149-188) | Pass Target: $\geq 90\%$ AviationExam

1. TERRESTRIAL MAGNETISM, VARIATION & DIP ANGLE

The Earth behaves like a giant bar magnet with its magnetic South pole located near geographic North. Aircraft magnetic navigation relies on three fundamental terrestrial parameters:

- **True North vs Magnetic North:** True North is the geographic rotational axis of the Earth. Magnetic North is the direction indicated by a freely suspended magnetic needle aligned with terrestrial flux lines.
- **Magnetic Variation (Declination):** The angular difference between True North and Magnetic North. Varies geographically: Variation West (Magnetic is West of True: $\text{True} = \text{Mag} - \text{Var}$); Variation East (Magnetic is East of True: $\text{True} = \text{Mag} + \text{Var}$). Mnemonic: 'East is Least (-), West is Best (+)'.
- **Isogonals & Agonic Line:** Isogonals are lines joining points of equal magnetic variation. The Agonic line is the line of ZERO variation.
- **Magnetic Dip (Inclination):** The angle between the Earth's magnetic flux lines and the horizontal local plane. At the Magnetic Equator (Aclinic line), $\text{Dip} = 0^\circ$ (purely horizontal). At the Magnetic Poles, $\text{Dip} = 90^\circ$ (purely vertical). Magnetic compasses become completely useless near the magnetic poles!

2. DIRECT-READING COMPASS TURNING & ACCELERATION ERRORS

Direct-reading magnetic compasses use a pendulously suspended magnet system whose center of gravity lies below the pivot point. Due to magnetic dip, the compass suffers severe turning and acceleration errors in the Northern Hemisphere:

Compass Error Scenario	Physical Mechanism (Inertia + Magnetic Dip)	Pilot Operational Correction Rule
Acceleration Errors (East/West Headings)	Accelerate $\rightarrow$ Inertia holds weight back $\rightarrow$ Card tilts $\rightarrow$ Dip pulls needle NORTH! Decelerate $\rightarrow$ Card tilts opposite $\rightarrow$ Needle pulls SOUTH.	Mnemonic: ANDS (Accelerate North, Decelerate South). On East or West headings in Northern Hemisphere!
Turning Errors (North/South Headings)	Turning THROUGH North: Centrifugal force tilts card outwards; dip pulls needle in turn direction $\rightarrow$ Compass LAGS or turns wrong way! Turning THROUGH South: Dip pulls needle ahead of turn $\rightarrow$ Compass LEADS (spins faster than aircraft).	Mnemonic: UNOS (Undershoot North, Overshoot South). Roll out BEFORE reaching North; roll out PAST South!

[!] EXAM TRAP (AVIATIONEXAM) - Northern Hemisphere Compass Errors Rules

- ANDS (Acceleration): When accelerating on East or West heading $\rightarrow$ Compass indicates a turn towards NORTH! When decelerating $\rightarrow$ Compass indicates a turn towards SOUTH! - UNOS (Turning): When turning to a Northerly heading $\rightarrow$ Undershoot roll-out (roll out $\sim 20\text{-}30^\circ$ before heading). When turning to a Southerly heading $\rightarrow$ Overshoot roll-out (roll out $\sim 20\text{-}30^\circ$ past heading)! - In the Southern Hemisphere: Acceleration and turning errors are EXACTLY REVERSED (SAND / ONUS)!

3. AIRCRAFT DEVIATION & COMPASS SWING

Aircraft metals and electrical wiring generate local magnetic fields that deflect the compass needle away from Magnetic North to Compass North:

- Deviation: The angular difference between Compass Heading and Magnetic Heading: Magnetic = Compass +/- Deviation.
- Compass Swing: A calibrated ground maintenance procedure conducted on a certified compass rose to align permanent corrector magnets, minimizing residual deviation. Residual errors are recorded on the flight deck Compass Deviation Card (maximum residual deviation $\leq 5^\circ$).

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): ANDS and UNOS Compass Error

Rule ANDS (East/West Headings): - Accelerate -> Compass turns NORTH. - Decelerate -> Compass turns SOUTH. Rule UNOS (Turning Headings in Northern Hemisphere): - Turning to NORTH: Compass leads/skips ahead -> Roll out EARLY (Undershoot North). - Turning to SOUTH: Compass lags behind -> Roll out LATE (Overshoot South).

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: In the Northern Hemisphere, what does a direct-reading magnetic compass indicate when the aircraft ACC

- [A]: A turn towards South
- [B]: A turn towards NORTH
- [C]: No change in heading
- [D]: Rapid 360-degree rotation

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Mnemonic ANDS: Accelerate North, Decelerate South. Accelerating on an East or West heading tilts the pendulous card, allowing the vertical component of Earth's magnetic field (dip) to deflect the card towards North.

Q2: When making a turn onto a heading of NORTH (360°) in the Northern Hemisphere, how should the pilot roll

- [A]: Exactly when the compass passes 360°
- [B]: Roll out BEFORE reaching 360° (Undershoot North), because the compass lags behind the turn
- [C]: Roll out past 360° (Overshoot North)
- [D]: Increase turn rate to Rate 2

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Mnemonic UNOS: Undershoot North, Overshoot South. When turning through North in the Northern Hemisphere, compass liquid drag and dip cause the card to lag behind the turn, requiring an early rollout.

Q3: What is the name of the line on an aeronautical chart connecting points of ZERO magnetic variation?

- [A]: Isogonic line
- [B]: Agonic line
- [C]: Aclinic line
- [D]: Rhumb line

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. An isogonal connects points of equal variation; the Agonic line is the specific isogonal of zero magnetic variation (where True North and Magnetic North coincide).